TFPT — Appendix H: the horizon unit system

One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code
(Hawking, de Sitter, Unruh, Page, scrambling — *standard physics in seam units, not new gravity*)

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What this document is about

A change of bookkeeping, not new gravitational physics: if gravity is the geometry-channel readout of the seam, then *all* horizons read the same boundary constant $c_3 = \frac{1}{8\pi}$. This note collects the readouts — Hawking, de Sitter and Unruh temperature, black-hole thermodynamics, the Page time, scrambling, the Nariai bound, $v_{\mathrm{GW}} = c$ and cosmic birefringence — in seam units, with two genuine compiler fingerprints ($1920 = |W(D_5)|$, $|\mu_4| = 4$). The Hawking normalisation itself is since *measured* on the seam: the reconstructed free OS quotient is KMS with $\beta_{\mathrm{angle}} = 2\pi$ exact, $T_{\mathrm{seam}} = 4c_3$ ([`verification/v526_seam_thermal_kms_nariai_bridge.py`]) — temperature joins geometry and anomaly as the *third leg* of c_3 ([C]-typed reading; the entropy-fraction bridge honestly does not close).

Contents

1 The seam constant is the universal horizon temperature factor 4

2 Schwarzschild thermodynamics in four c_3-lines 4

3 Page time and scrambling 5

4 De Sitter and the Nariai bound from the Λ closure 5

5 The maximal black hole is the anchor: SdS in seam units 6

6 Turnaround radius, gravitational waves, light 14

7 The local sibling: an EHT achromatic polarization intercept 15

8 Search targets (not claims) 15

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\mathrm{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions** (plus one standalone working note), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to,

and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

Working notes: N1. `note_e8_gaussian_code` — the Gaussian code bridge: E_8 over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8,4,4]$, the canonical four-bit quotient $L/(1+i)L \cong \mathbb{F}_2^4$, and the G_{31} quartic companion (exact algebra, machine-certified).

You are reading companion H (Appendix H — Horizon Readouts).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1,1,2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725)	closed metrology [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Beside that compiler rest sits the strict-physical-TOE rest: the ten-component Rest_{TOE} / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}} := \text{AND}(\text{T1..T8})$; not a closure).

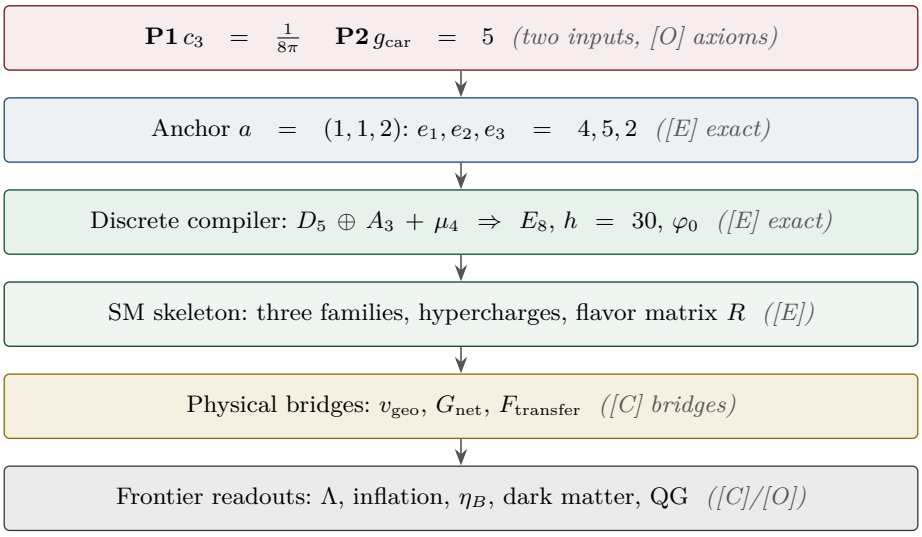
The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD $= 1$ (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print`

`axioms` check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package (h_s=16/16=1∈ℤ, the Longo–Rehren locality criterion; Böckenhauer; KLM μ=4/2²=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed'` (`v469`); so the MMST route is *closed modulo cited theorems*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin 1.648 > 0, `v76`/`v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test



Scope and honest typing. If gravity is the geometry-channel readout of the seam (companion note "Seam response grammar"), then horizons are not separate miracles — they all read the *same* boundary normaliser $c_3 = 1/(8\pi)$. This note collects the readouts. Markers: [C] is standard horizon physics rewritten in TFPT/seam units (no new claim) or a search target; [E] is an exact compiler connection (audit-level) or a reproduced numerical readout; [O] is a search target, not a result. Nothing here is sold as new gravitational physics; it is a *change of bookkeeping* that exposes shared structure.

Unit conventions used throughout

Three unit systems appear; the seam factor $c_3 = 1/(8\pi)$ is the same number in all of them, only the carried dimensionful constants change:

system	convention	Hawking temperature
Planck	$G{=}\hbar{=}c{=}k_B{=}1$	$T_H = c_3/M = 1/(8\pi M)$
reduced Planck	$\bar{M}_{\rm Pl} = M_{\rm Pl}/\sqrt{8\pi}$, $\hbar{=}c{=}k_B{=}1$	$T_H = c_3\,\bar{M}_{\rm Pl}^2/M$
SI	restore $G, \hbar, c, k_B$	$T_H = \dfrac{\hbar c^3}{8\pi G k_B M} = c_3\,\dfrac{\hbar c^3}{G k_B M}$

All boxed formulas below are in Planck units unless stated; the seam constant is dimensionless.

Standard references for the formulas reframed here: Hawking temperature and radiation (Hawking 1975); Bekenstein–Hawking entropy (Bekenstein 1973, Hawking 1975); Page time (Page 1993); fast scrambling $t_{\rm scr} \sim \beta \log S$ (Sekino–Susskind 2008, Maldacena–Shenker–Stanford 2016); Nariai bound (Nariai 1950); de Sitter thermodynamics (Gibbons–Hawking 1977). *Nothing below is a new derivation of these* — only a rewriting in the seam unit c_3.

1 The seam constant is the universal horizon temperature factor

Every Killing horizon has $T = \hbar\kappa/(2\pi c k_B)$. Since

$$\boxed{\frac{1}{2\pi} = 4c_3, \qquad \frac{1}{8\pi} = c_3} \qquad \textbf{[E]},$$

the universal factor *is* the seam constant: $T_{\rm hor} = 4c_3\,\hbar\kappa/(ck_B)$. Black holes, de Sitter and Unruh therefore share *one* thermal grammar. **[C]** [verification/v8_horizon.py]

2 Schwarzschild thermodynamics in four c_3-lines

For a non-rotating, uncharged hole in Planck units ($G{=}\hbar{=}c{=}k_B{=}1$):

$$\boxed{T_H = \frac{c_3}{M}, \qquad S_{BH} = \frac{M^2}{2c_3}, \qquad P_H = \frac{c_3}{1920\,M^2}, \qquad \tau_{\rm evap} = \frac{640}{c_3}M^3} \qquad \textbf{[C]}.$$

Two of the constants are compiler numbers, not coincidences:

- the Hawking temperature factor is $c_3 = 1/(8\pi)$ — the same P1 boundary normaliser; **[E]**
- the radiated-power denominator is $1920 = |W(D_5)|$, the Weyl-group order of the D_5 carrier. **[E]**

Scope of P_H: this is the idealised massless single-channel blackbody normalisation. Greybody factors and the radiated species content shift the phenomenological evaporation coefficient; the c_3-form is the convention statement, not a general phenomenological prediction.

object	T_H	lifetime / entropy
solar-mass hole	6.17×10^{-8} K	$\tau_{\rm evap} \approx 2.1 \times 10^{67}$ yr; $S/k_B \approx 1.05 \times 10^{77}$
PBH evaporating now (1.7×10^{11} kg)	$k_B T \approx 61$ MeV	relevant for γ/ν PBH signatures

The temperature is modular; the entropy carries the scalaron. [verification/v208_bh_thermodynamics.py] The Hawking factor is not put in by hand: the stationary exterior modular flow $\Delta^{it} = e^{-2\pi t K_H}$ (Bisognano–Wichmann / Tomita–Takesaki) makes the outside state KMS at inverse temperature 2π, so $T_H = \kappa/(2\pi)$ — and that 2π *is* the seam unit $1/(4c_3)$, reproducing $T_H = c_3/M$

above (the seam=horizon modular identification ties to $[\rho, \Lambda_\Sigma] = 0$, [verification/v198_modular_commutator_reduction.py]). The same normalisation is now *measured* on the reconstructed free seam OS quotient (`SEAM.THERMAL.KMS.01`, [verification/v526_seam_thermal_kms_nariai_bridge.py]): detailed balance gives $\beta = N$ clock steps at both levels, hence $\beta_{\text{angle}} = 2\pi$ and $T_{\text{seam}} = 4c_3$ — temperature joins geometry and anomaly as the *third leg* of c_3 ([C]-typed reading: seam euclidean circle = thermal circle of the reconstructed horizon dynamics; the entropy-fraction bridge to the Nariai ledger honestly does not close). The induced $R + R^2$ scalaron then corrects the area law to the Wald entropy

$$S_W = \frac{f_R\, A}{4G} = \frac{A}{4G}\Big(1 + \frac{R_h}{3M_s^2}\Big), \qquad M_s = c_3^{7/2}\bar{M}_{\text{Pl}},$$

an *exact* consequence of $f(R) = R + R^2/(6M_s^2)$ ([verification/v28_gravity_fR.py]); the leading $A/4G$ is the c_3 area law ($1/4 = 1/|\mu_4|$). [E] for the identities; the black-hole/modular identification is [C].

3 Page time and scrambling

Since $S_{BH} \propto M^2$ and $\tau \propto M^3$, the Page point ($M \to M/\sqrt{2}$) is

$$t_{\text{Page}} = \Big(1 - \tfrac{1}{2\sqrt{2}}\Big)\tau_{\text{evap}} \approx 0.6464\,\tau_{\text{evap}} \qquad \text{[C]}.$$

In TFPT the Page curve is *boundary recoverability*: the physical sector is OS-reconstructed and gapped, and the recovered mutual information decays at the transfer rate, $I(A{:}C|B)_n \sim \lambda_2^n = (2/3)^{6n}$. **This is the same operator that builds the Standard Model:** the boundary transport spectrum is $\{1, (2/3)^6, (1/3)^6\}$ and its sub-leading eigenvalue $\lambda_2 = (2/3)^6$ is exactly the flavor-gap eigenvalue ($\Delta_{\text{gap}} = -\log(2/3)^6 = 6\log\frac{3}{2}$). One boundary transport thus governs *both* the SM flavor hierarchy and the horizon information recovery. [E] [verification/v54_seam_horizon_keystones.py]

The recovery is a finite code ([verification/v221_seam_qecc.py]). Read as a quantum channel, this is a finite recoverability (error-correction-style) code: the code space is the carrier half-spinor $\dim S^+ = 16$, the gapped transport T on the cusp-weight 3-space (deviation directions $(1,-1,0)$ and the Nariai anchor $(1,1,-2)$) is symmetric and doubly stochastic (CPTP), and on the trace-zero subspace it contracts with operator norm exactly $(2/3)^6$, so $\|T^n\delta\| \le (2/3)^{6n}\|\delta\|$ — a Knill–Laflamme-type per-step recovery bound, with a free ratio breaking it. "Finite recoverability code" is [E]; the explicit Petz recovery map and the rank-one (one-dimensional baby-universe, cf. Engelhardt 2025) contraction limit $\|T^n - P_\infty\| = (2/3)^{6n}$ are now realised numerically (`experiments/recovery-channel`), so the Petz identification of T with the Page recovery map is verified at the channel level [C]; the full holographic reading stays gated on the Seam–Horizon theorem. The scrambling prefactor carries a μ_4 signature: with $\beta = 1/T_H = 8\pi M$, $\beta/2\pi = 4M$, so

$$\boxed{t_{\text{scr}} \sim |\mu_4|\, M \log S} \qquad (|\mu_4| = 4) \qquad \text{[E]-audit.}$$

4 De Sitter and the Nariai bound from the Λ closure

With the action-grammar Hubble/Λ readouts ($H_0/\bar{M}_{\text{Pl}} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$, $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 \sim e^{-2\alpha_\star^{-1}}$), the asymptotic de Sitter horizon gives

$$\boxed{\frac{T_{dS}}{\bar{M}_{\text{Pl}}} = 16c_3^2\, e^{-\alpha_\star^{-1}}, \qquad S_{dS} = \frac{e^{2\alpha_\star^{-1}}}{128c_3^4} = 32\pi^4 e^{2\alpha_\star^{-1}} \approx 3.32 \times 10^{122}} \qquad \text{[C]}.$$

This is the area-law horizon capacity of the universe from $(\alpha_\star^{-1}, c_3, H_\Lambda)$ alone. The maximal black hole in de Sitter (Nariai) is

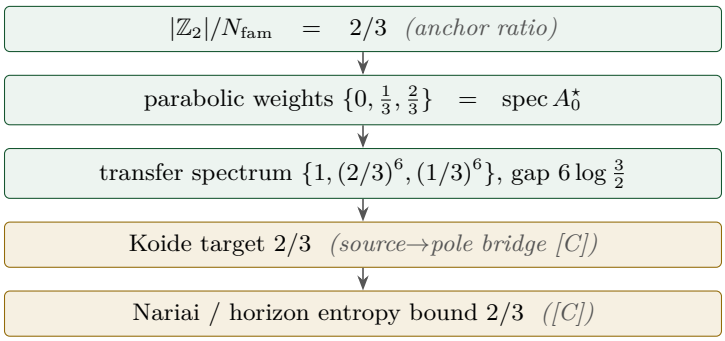
$$M_N \approx 2.16 \times 10^{22}\, M_\odot \approx 4.3 \times 10^{52}\,\text{kg}, \qquad S_N = \tfrac{1}{3}S_{dS} \qquad \text{[C]},$$

the absolute gravitational-capacity limit, not an observed structure.

5 The maximal black hole is the anchor: SdS in seam units

The Nariai limit is not just a capacity number — written in compiler units, the entire Schwarzschild–de Sitter (SdS) family lands on atoms that are already load-bearing elsewhere in TFPT, with no adjustable parameter on either side. **[E]** [verification/v101_horizon_anchor.py]

The defect reading: no singularity, a seam fixed point ([verification/v209_bh_regular_core.py]). In compiler units the interior is *not* a curvature blow-up but the seam attractor: the same gapped transport whose sub-leading eigenvalue is $\lambda_2 = (2/3)^6$ ([verification/v56_unique_attractor.py]) drives $\varphi \to \varphi_\star$ ($\mathrm{d}\varphi/\mathrm{d}t = 0$), so "$\rho \to \infty$" is replaced by a fixed point; information returns through the same Page-recovery channel (λ_2 per step, [verification/v54_seam_horizon_keystones.py]); and the end state is a holographic Planck-scale floor ($S_{BH} = A/4$, one cell per $|\mu_4| = 4$ Planck areas), not a point. *Honest:* a **[C]** structural reading — the old RN/torsion-charge metric is *not* resurrected, it is superseded by the Nariai/seam=horizon anchor here.



Green: exact motif ([E]). Gold: physical bridge ([C]) — the same ratio, two epistemic types.

Nariai = the anchor; the Koide 2/3 is the entropy bound [E] ([verification/v101_horizon_anchor.py])

With $f(r) = 1 - 2M/r - \Lambda r^2/3$, the horizon cubic at the maximal mass $M_N = 1/(3\sqrt{\Lambda}) = 1/(N_{\mathrm{fam}}\sqrt{\Lambda})$ is, in units $1/\sqrt{\Lambda}$,

$$\boxed{t^3 - 3t + 2 = (t-1)^2(t+2)} \qquad \text{roots } (1, 1, -2),$$

with coefficients $(-3, 2) = (-N_{\mathrm{fam}}, |\mathbb{Z}_2|)$ — and the root pattern is exactly the **traceless projection of the anchor**: $a - \frac{p_1(a)}{3}\mathbf{1} = -\frac{1}{3}(1, 1, -2)$ for $a = (1, 1, 2)$ (compare $\chi_a = (t-1)^2(t-2)$, the anchor characteristic polynomial). Five exact companions:

- **Entropy bound = the Koide branch value.** Each Nariai horizon carries $S_{dS}/3 = S_{dS}/N_{\mathrm{fam}}$ and the total is $\boxed{S_{\mathrm{Nariai}}/S_{dS} = 2/3 = |\mathbb{Z}_2|/N_{\mathrm{fam}}}$ (deficit exactly $S_{dS}/3$). The interpolation in $x = r_b/r_c$ is the closed form $S_{\mathrm{tot}}/S_{dS} = (x^2{+}1)/(x^2{+}x{+}1)$ — denominator $= \Phi_3(x)$, the N_{fam} cyclotomic — monotone from 1 to 2/3 with its minimum exactly at the merge (Fig. 2a). The floor admits a *variation-free* one-line proof: $3(x^2{+}1) - 2\Phi_3(x) = (x{-}1)^2 \geq 0$, so $S_{\mathrm{tot}} \geq \frac{2}{3}S_{dS}$ with equality *iff* $x = 1$ (the Nariai merge) **[E]** ([verification/v190_nariai_entropy_bound.py]).
- **Three-sheet conservation.** The cubic is traceless with fixed e_2, so $\pi \sum_i r_i^2 = 6\pi/\Lambda = |\mathbb{Z}_2|\, S_{dS}$ for *every* M: the black hole redistributes entropy between the two physical sheets and the virtual third root; the three-sheet total never moves.
- **Koide-form quotient of the roots.** $Q_{\mathrm{geom}} = \sum r_i^2/(\sum |r_i|)^2$ runs monotonically over $[\frac{3}{8}, \frac{1}{2}]$: pure dS gives $\frac{1}{2} = |\mathbb{Z}_2|/|\mu_4| = \delta$ (the half-step), Nariai gives $\frac{3}{8} = p_2(a)/e_1(a)^2$ — exactly the two nonzero $SU(4)_1$ conformal weights (Fig. 2b).
- **The mass line is a double cover.** $\mathrm{disc}_r \propto (1 - 3m)(1 + 3m)$ in $m = M\sqrt{\Lambda}$: branch points $m = \pm 1/N_{\mathrm{fam}}$, separation $2/3 = |\mathbb{Z}_2|/N_{\mathrm{fam}}$, deck involution = the black-hole ↔ cosmological horizon swap — the gravitational twin of the flavor sheet $\mathbb{Z}_2$ and of the split N_{fam}-linear cover $(3x{+}2)(3x{+}5)$ of the flavor sector (Fig. 1). The two branch structures share one N_{fam}-linear ramification: the affine map $q = \frac{7}{2} + \frac{N_{\mathrm{fam}}^2}{2}m$ carries the SdS branch points $m = \pm 1/N_{\mathrm{fam}}$ onto the flavor branch labels $\{2, 5\} = \{|\mathbb{Z}_2|, g_{\mathrm{car}}\}$ (midpoint 7/2). This is an exact *relabeling* — the shared invariant is the 2/3 ramification, not the map, which exists for any two pairs of points — so it is recorded as

a [C] structural alignment, not a theorem ([verification/v191_universal_branch_line.py]).

- **Temperature lemma / orientation.** $|\kappa_b/\kappa_c| = (2x{+}1)/(x(x{+}2)) > 1$ for $x < 1$ (equality iff $x = 1$): the black-hole sheet is *always* hotter, so evaporation flows *away* from the merge — the anchor configuration is the **repeller** and the democratic endpoint the attractor, the same orientation as the flavor flow (Koide attractor / carrier repeller, the frontier note). *Audit reading.*

Completing the seam-unit table: $dM = c_3\,\kappa\,dA$ (first law), $M = 2c_3\,\kappa A$ (Smarr), $S \le ER/(4c_3)$ (Bekenstein), $P_H = c_3/(1920\,M^2)$, lifetime $\tau = 5120\pi M^3 = 128\,g_{\rm car}M^3/c_3$ ($128 = 2^7$ the dS prefactor, 7 the scalaron exponent; photon/geometric-optics conventions, typed external), Kerr $A_{\rm ext} = M^2/c_3$ with $A_{\rm ext}/A_{\rm Schw} = 1/2 = 1/|\mathbb{Z}_2|$ and $S_{\rm ext} = M^2/(4c_3)$, and the Bekenstein–Mukhanov–Hod area quantum $4\ln 3 = \ln 81 = \ln(N_{\rm fam}^4) = \ln(\text{disc of the flavor cover})$ ([C] as the QNM reading). *Honest scope:* the GR identities are textbook facts; the [E] content is six independent landings on already-load-bearing atoms with zero free parameters. The "carrier in the bulk" reading (two glue chiralities = two horizon sheets, ramification = Nariai) stays [C]. No new data test (M_N today $\sim 10^{22}M_\odot$): a structure surface, not a measurement surface.

The dual anchor: the inverse flavor response *is* the Nariai root [E] ([verification/v134_dual_anchor.py])

The Nariai pattern is not only the traceless anchor — it is stored *inside the flavor compiler* as a dual invariant. For the residue/winding operators R, L of the flavor sector and the anchor $a = (1,1,2)$,

$$\boxed{d := a^\top R^{-1} = a^\top L^{-1} = \left(-\tfrac{1}{2}, -\tfrac{1}{2}, 1\right)}, \qquad d\cdot\mathbf{1} = 0, \quad d\cdot a = 1, \quad (1,1,-2) = -2d,$$

i.e. $d = \frac{3}{2}\left(a - \frac{4}{3}\mathbf{1}\right)$ is the *normalised traceless anchor*, proportional to the Nariai double-root pattern above. The invariance is exact and structural (Sherman–Morrison): with $L = R + 6\,\mathbf{1}e_1^\top$, a covector is winding-invariant iff it annihilates $R^{-1}\mathbf{1} = \frac{1}{4}(1,1,-1)$ — the winding deformation lives in the democratic direction $\mathbf{1}$, which d kills. Controls: $\mathbf{1}$, e_1 and the torsion normal $n = (5,-9,6)$ do *not* lie on the invariance plane $(\frac{1}{4}, \frac{1}{4}, -\frac{5}{2})$, so the anchor's membership is special. Together (d, n) form the dual normal pair of the flavor boundary: d reads the traceless horizon structure, n reads first-generation torsion ($n^\top R = (8,0,0)$, $n^\top L = (20,0,0)$). *Typing:* all identities [E]; the reading — a third, purely *algebraic* leg of the flavor ↔ horizon bridge, beside the shared clock spectrum ([verification/v126_clock_wall_bridge.py]) and the entropy power law ([verification/v129_entropy_power_law.py]) — stays [C].

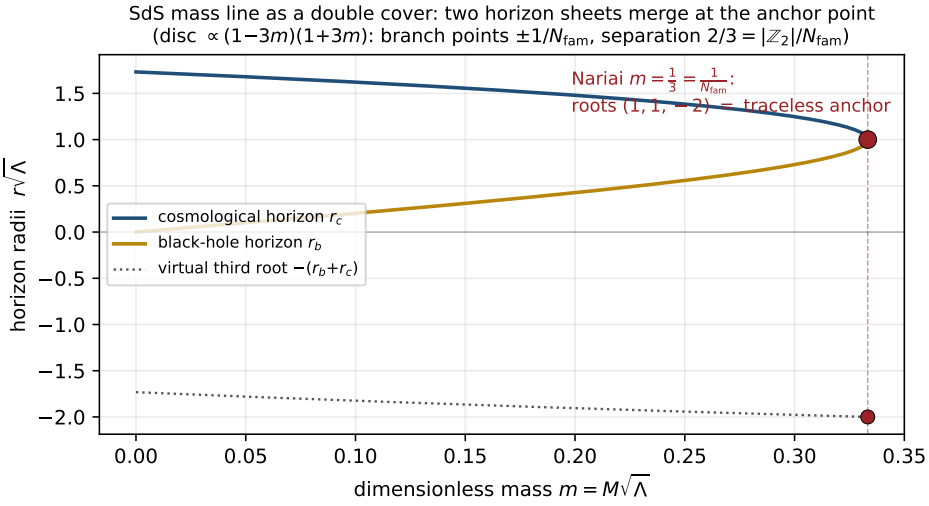


Figure 1: The SdS mass line as a double cover: black-hole and cosmological horizon sheets merge at the Nariai point $m = 1/N_{\rm fam}$, where the three roots form the traceless anchor pattern $(1,1,-2)$. [verification/v101_horizon_anchor.py]

One orientation: the anchor is the *stationary repeller* in both sectors ([E]/[C]). The evaporation direction of the temperature lemma upgrades to exact variational structure on both

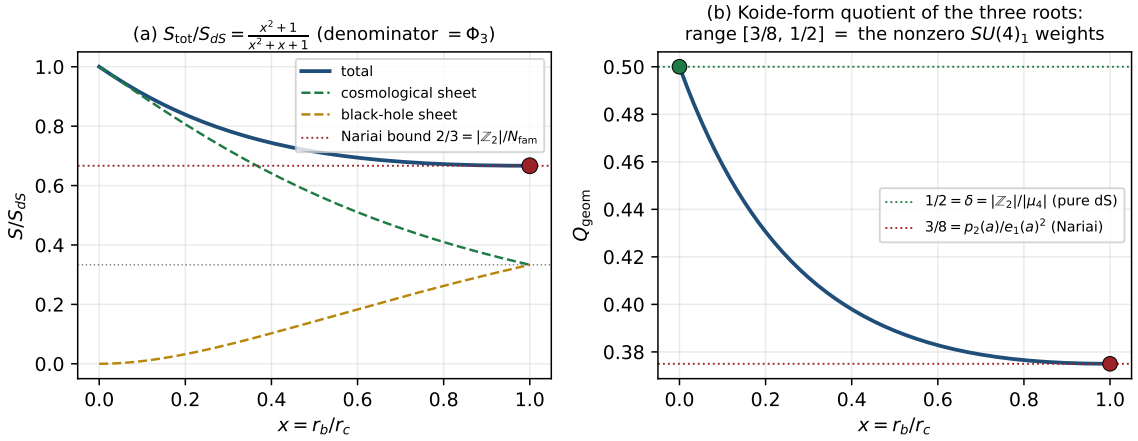


Figure 2: (a) Total SdS entropy interpolates from S_{dS} to the Nariai bound $\frac{2}{3}S_{dS}$ via $(x^2{+}1)/\Phi_3(x)$. (b) The Koide-form quotient of the three roots runs exactly over the nonzero $SU(4)_1$ weights $[\frac{3}{8}, \frac{1}{2}]$. [verification/v101_horizon_anchor.py]

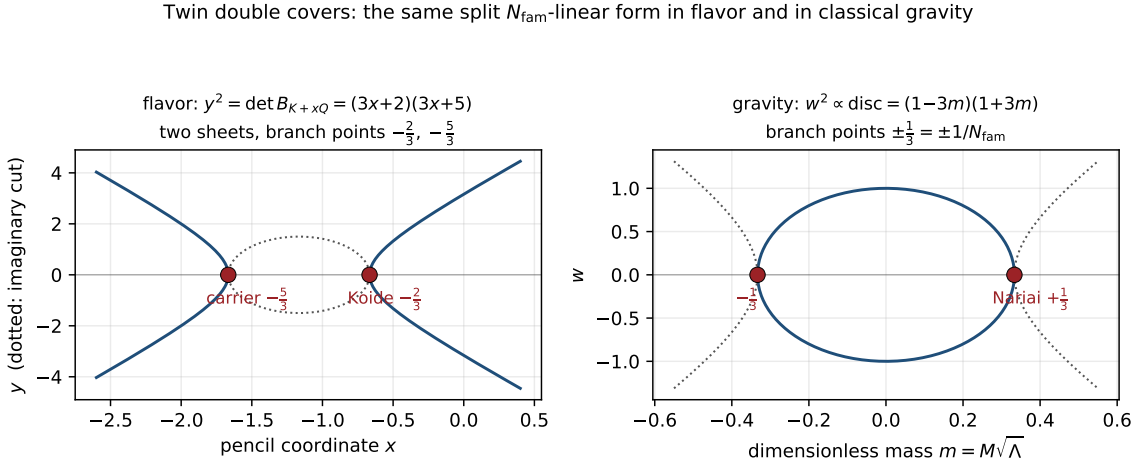


Figure 3: Twin double covers: the flavor anchor-block cover $y^2 = (3x{+}2)(3x{+}5)$ and the SdS mass-line cover $w^2 \propto (1{-}3m)(1{+}3m)$ share the split N_{fam}-linear form with rational branch points at spine-over-family values. [verification/v101_horizon_anchor.py]

one-dimensional moduli lines (Fig. 4). **Flavor:** the canonical relaxation $\dot{q} = (\Delta/N_{\mathrm{fam}})(q{-}2)(q{-}5)$ is the *gradient flow* of the cubic potential

$$V(q) = -\frac{\Delta}{N_{\mathrm{fam}}}\Big(\frac{q^3}{3} - \frac{7}{2}q^2 + 10q\Big), \qquad \boxed{V''(2) = +\Delta, \quad V''(5) = -\Delta}$$

— the critical points are exactly the two branch points, the stationary curvatures are ±the transfer gap, the inflection sits at $q = \frac{7}{2}$ = scalaron/2, and the Lyapunov rate is constant: $d(-\ln\rho)/dt = \Delta$ exactly. **Gravity:** the SdS entropy functional has

$$\frac{d}{dx}\frac{S_{\mathrm{tot}}}{S_{dS}} = \frac{(x-1)(x+1)}{\Phi_3(x)^2}, \qquad \boxed{\Big(\frac{S_{\mathrm{tot}}}{S_{dS}}\Big)''(1) = \frac{2}{9} = \frac{|\mathbb{Z}_2|}{N_{\mathrm{fam}}^2} = \frac{2}{3}\cdot\frac{1}{3}}$$

— the Nariai/anchor point is the *unique* physical stationary point, its curvature is a grammar constant (= the product of the two Nariai entropy fractions), and the temperature lemma makes the evaporation an entropy *ascent* away from it. **Orientation theorem [E]:** in both sectors the anchor-type configuration is a stationary point of the natural functional and the dynamics

flows away from it, with grammar-constant stationary curvatures. *Honest disanalogies (recorded):* the flavor attractor is itself a branch point while the gravity attractor is the smooth endpoint; the flavor functional is the potential of a [C]-typed relaxation while the gravity functional is the Bekenstein–Hawking entropy — "one variational principle of the seam" stays a *reading* [C], and no open gate is touched. [verification/v102_seam_orientation.py]

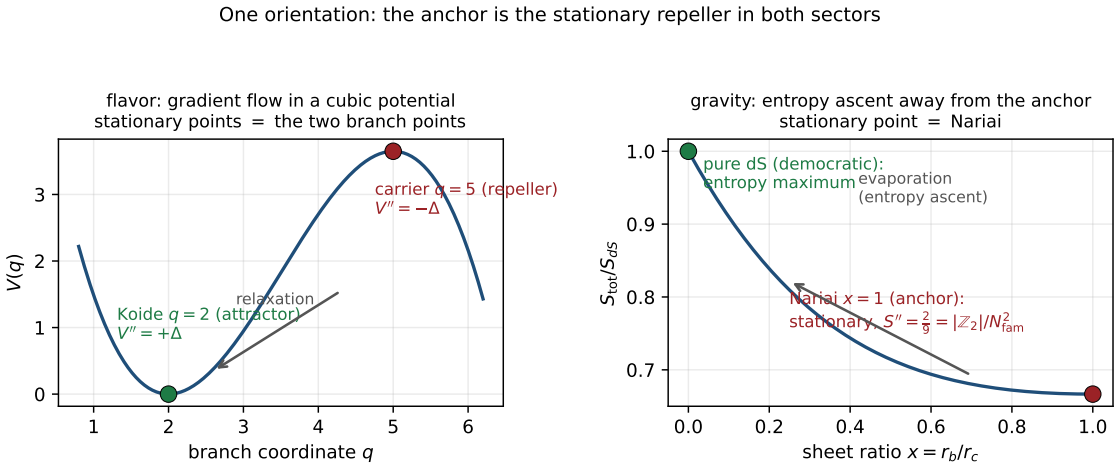


Figure 4: The orientation theorem: the anchor configuration is the stationary repeller of the natural functional in both sectors — flavor (cubic potential, curvatures $\pm\Delta$) and gravity (SdS entropy, curvature $2/9 = |\mathbb{Z}_2|/N_{\rm fam}^2$). [verification/v102_seam_orientation.py]

The trisection normal form: the canonical coordinate exists [E] ([verification/v103_trisection_normal_form.py])

The curvature-matching question of the previous box has a closed answer. The SdS horizon cubic is uniformized by **angle trisection**: with $r = 2\cos\theta$ (units $1/\sqrt{\Lambda}$), $r^3 - 3r = 2\cos 3\theta$ *exactly*, so the cubic is $\cos 3\theta = -3m$ — the three horizon roots are the three trisection branches (the $\mathbb{Z}_3$ rotation = the triality deck, cf. $\operatorname{coker} Q = \mathbb{Z}/N_{\rm fam}$). In the centered angle ψ ($\cos\varphi = -3m$, $\psi = \varphi - \pi$; Nariai $\psi = 0$, pure dS $\psi = \pi/2$, deck $\psi \to -\psi$):

$$m = \frac{\cos\psi}{N_{\rm fam}}, \qquad \boxed{\frac{S_{\rm tot}}{S_{dS}} = \frac{4}{3} - \frac{2}{3}\cos\frac{2\psi}{3}}$$

— *one* cosine whose mean ($|\mu_4|/N_{\rm fam}$), amplitude and frequency (both $|\mathbb{Z}_2|/N_{\rm fam}$) are glue atoms over the family count (Fig. 5). Three exact consequences:

- **Canonical curvature = the Koide constant to the family power:** $\sigma''(\psi{=}0) = \frac{8}{27} = (\frac{2}{3})^3 = (|\mathbb{Z}_2|/N_{\rm fam})^{N_{\rm fam}}$.
- **Invariant base slope:** $d\sigma/dm|_{\rm Nariai} = -\frac{8}{9} = -\operatorname{rank} E_8/N_{\rm fam}^2$ (coordinate-invariant, cross-checked in both coordinates; generally $-\frac{4}{3}\sin(2\psi/3)/\sin\psi$); dimensionful $dS_{\rm tot}/dM|_N = -r_N/(N_{\rm fam}c_3)$.
- **The bridge:** the flavor invariant is a *rate*, multiplier $(2/3)^6 = (2/3)^{2N_{\rm fam}}$ per transport step; the gravity invariant is a *curvature*, $(2/3)^{N_{\rm fam}}$ in the canonical angle — same base $2/3 = |\mathbb{Z}_2|/N_{\rm fam}$, exponent ratio $2 = |\mathbb{Z}_2|$. What is still missing for a full identification is the gravity-side *clock* (the near-Nariai evaporation generator that converts curvature into a rate) — the same class of missing object as the Koide transfer generator (frontier note): **one clock, two known geometries**. That identification stays [C].

The classical clock speaks anchor — and the honest $(2/3)$-test ([E]/[C]). The clock question of the previous box has a *classical* half, and it is pure GR (Ginsparg–Perry 1983; Bousso–Hawking class): linearize around the Nariai geometry $dS_2 \times S^2$. **Static pin [E]:** on the dS_2 static patch the function $\phi(\rho) = \rho$ satisfies $\frac{d}{d\rho}[(1-\Lambda\rho^2)\phi'] = -2\Lambda\phi$ *exactly* — and this static mode *is* the exact SdS

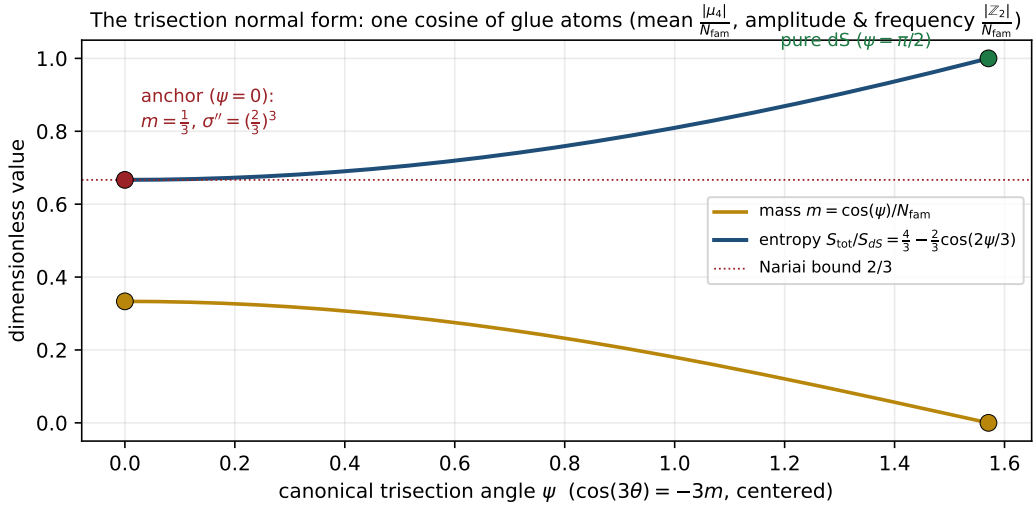


Figure 5: The trisection normal form: in the canonical angle ψ the mass and the total entropy are pure (co)sines built from glue atoms; the anchor sits at $\psi = 0$ with canonical curvature $(2/3)^3$. [verification/v103_trisection_normal_form.py]

two-horizon deformation, so the S^2-radius modulus mass is pinned internally: $m^2 = -2\Lambda = -|\mathbb{Z}_2|\Lambda$ (the Laplace-type form of the linearization is the one external GR import, typed honestly). The Euclidean spectrum is the Ginsparg–Perry tower $(l(l+1)-2)\Lambda$: exactly *one* negative mode $-|\mathbb{Z}_2|\Lambda$. **The clock [E]:** in Hubble units the homogeneous modes $\phi \sim e^{\lambda H t}$ obey

$$\boxed{\chi_{\rm clock}(\lambda) = \lambda^2 + \lambda - 2 = (\lambda - 1)(\lambda + 2)}$$

— **the anchor quadratic**: eigenvalues $\{1, -2\}$ = the distinct anchor roots, and the Nariai cubic factors as $(t-1)^2(t+2) = (t-1)\cdot\chi_{\rm clock}(t)$. The anchor thus appears a *third* time: as configuration roots, as the curvature base, and as the spectrum of the linearized clock. The horizon split is linear in the canonical angle with coefficient $1/\sqrt{N_{\rm fam}}$, and the entropy deviation grows at exactly $2H = |\mathbb{Z}_2|\times$Hubble. **Honest $(2/3)$-test [C]:** the classical clock eigenvalues are *integers*, not $(2/3)$-powers — the $(2/3)$-powers stay in the functional (curvature $(2/3)^{N_{\rm fam}}$). The conversion curvature→rate at one loop (quantum (anti-)evaporation) remains external QFT input: the bridge to the flavor rate $(2/3)^{2N_{\rm fam}}$ is *not* closed here. The seam variational principle now has exactly *one* missing object: the quantum clock. [verification/v104_nariai_clock.py]

The quantum-clock target, made quantitative ([E]/[C]). Instead of guessing the quantum clock, its job description can be pinned exactly. **(1) The classical clock already lives on the cusp ladder [E]:** across the physical Ginsparg–Perry modes ($l = 0$: $(\lambda{-}1)(\lambda{+}2)$; $l = 1$: $\lambda(\lambda{+}1)$) the decay exponents are

$$\boxed{\{0, -1, -2\} = -\,\mathrm{Spec}(V) = -N_{\rm fam} \times \text{cusp weights}}$$

— exactly the grading that carries the transfer operator's eigenspaces; the $l \geq 2$ modes are damped oscillations with $\mathrm{Re}\,\lambda = -\frac{1}{2} = -\delta$ (audit). **(2) The exact target [E]:** the transfer rates $\{0, \Delta, 6\ln 3\}$ are *not* weight-linear — rate(2)/rate(1) $= \log_{3/2} 3 = 1 + \log_{3/2} 2 = 2.7095$, with the exact identity $(1/3)^6 = \left((2/3)^6\right)^{\log_{3/2} 3}$: *the quantum clock must bend the weight-linear spectrum by* $\log_{9/4} 3 = 1.35476$ *per weight step.* **(3) Coupling landscape [E]:** the dimensionless clock coupling $\kappa = (c/24\pi)(\Lambda/\bar{M}_{\rm Pl}^2)$ with $c = 8$ is 7.6×10^{-122} at the cosmological Nariai (hopeless) and $\kappa = 1/(3\pi) = 0.106$ at seam curvature — the only regime where an $O(0.35)$ bend is reachable, and it is borderline non-perturbative: one-loop linear response cannot fix the bend; an $O(1)$ resummation or an exact seam construction is required. *Status [C]:* R1 is sharpened from "find a

clock" to "produce the bend $\log_{3/2} 3$ on the cusp ladder at coupling $O(1/3\pi)$". Nothing is fitted; the identification stays open. [verification/v107_quantum_clock_target.py]

The resummed clock: the bend has a closed form ([E]/[C]). The "bend" is not an exotic target — the frozen transfer spectrum *is* a resummed logarithm. **(1) Spectrum reading [E]:** the eigenvalues $\{1, (\frac{2}{3})^6, (\frac{1}{3})^6\}$ are exactly

$$\boxed{\lambda_n = \Big(1 - \frac{n}{N_{\rm fam}}\Big)^{p_2}, \qquad n = 0, 1, 2} \qquad (p_2 = 6 = |R^+(A_3)|),$$

so the clock has the closed form $\mathrm{rate}(n) = -p_2 \ln(1 - n/N_{\rm fam})$ — it reproduces $\{0, \Delta, 6\ln 3\}$, and the v107 bend $\log_{3/2} 3$ becomes a one-line consequence. **(2) The linearisation carries the sheet [E]:** $\mathrm{rate}(n) = |\mathbb{Z}_2|\, n + \frac{n^2}{3} + \frac{2n^3}{27} + \cdots$ — the one-loop term has slope exactly $|\mathbb{Z}_2| = 2$ relative to the *integer* classical Ginsparg–Perry clock $\{0, -1, -2\}$: the quantum clock is the sheet-doubled classical clock plus the geometric tail $\sum_k (n/N)^k/k$. **(3) The wall at $N_{\rm fam}$ [E]:** rate $\to \infty$ as $n \to N_{\rm fam}$ — the resummed logarithm has a pole at the family count, so the cusp ladder *must* end at $n = N_{\rm fam} - 1 = 2$: the three-level transfer spectrum is forced by the clock itself. *R1 re-sharpened [C]:* the semiclassical job is now "derive $-p_2 \ln(1 - n/N_{\rm fam})$ from the near-Nariai one-loop at $\kappa = 1/(3\pi)$" — a closed-form target, with the linear-response slope $|\mathbb{Z}_2|$ as its first checkpoint. All spectra frozen (v54/v69/v95/v104/v107); nothing fitted. [verification/v124_resummed_clock.py]

The clock and the wall share a spectrum ([E]/[C]). Both ends of the resummed clock connect to established exact objects. **(1) The weights are the parabolic weights [E]:** the arguments $n/N_{\rm fam} = \{0, \frac{1}{3}, \frac{2}{3}\}$ of the resummed logarithm are *exactly* $\mathrm{spec}\, A_0^\star$ — the spectrum of the exact $(U_{\rm wall})$ residue (research contracts):

$$\boxed{\lambda_n = (1 - \alpha_n)^{p_2}, \qquad \alpha_n \in \mathrm{spec}\, A_0^\star = \{0, \tfrac{1}{3}, \tfrac{2}{3}\}}$$

— the horizon clock and the flavor wall share their spectrum through the complement map $\alpha \mapsto 1 - \alpha$. **(2) Checkpoint 1 passes classically [E]:** the clock's linear coefficient $p_2/N_{\rm fam} = 2 = |\mathbb{Z}_2|$ *equals* the established classical entropy-deviation rate $2H = |\mathbb{Z}_2| \times$Hubble (v104) — the slope-2 linear response is already the classical entropy rate; the genuinely *quantum* content of R1 isolates to the geometric tail $\sum_{k\geq 2} \alpha^k/k$. **(3) The determinant is the entropy curvature [E]:** $\det(\mathbf{1} - A_0^\star) = \frac{2}{9} = |\mathbb{Z}_2|/N_{\rm fam}^2$ — exactly the gravity entropy curvature at the anchor (v102), a fourth independent appearance; $\mathrm{tr}(\mathbf{1} - A_0^\star) = 2 = |\mathbb{Z}_2|$, $\det A_0^\star = 0$. *R1 re-targeted [C]:* derive "rate $= -p_2 \ln(1 - \alpha)$" with α the Mehta–Seshadri parabolic weight — first order classical, tail quantum: **one spectrum, two gates, one object** $A_0^\star$. [verification/v126_clock_wall_bridge.py]

The clock is a log-determinant: the tail is the ring (RPA) resummation ([E]/[C]). The "$O(1)$ resummation" demanded by v107 is now *typed*. For a rank-one insertion P ($\mathrm{tr}\, P^k = 1$), $-\ln\det(\mathbf{1} - \alpha P) = -\ln(1 - \alpha) = \sum_k \alpha^k/k$, so the resummed clock is exactly

$$\boxed{\mathrm{rate} = -p_2\ \mathrm{Tr}\ln(\mathbf{1} - \alpha P)}$$

— $p_2 = 6$ *identical* one-loop ring (RPA) towers, **one per hexagon site** (= positive root of A_3), with coupling $\alpha =$ the parabolic weight. *Ring ladder [E]:* the k=1 ring is the *classical* term ($6\alpha = |\mathbb{Z}_2| n$, the v104/v126 entropy rate); the k=2 ring is the first quantum correction ($\frac{1}{3}$ at weight 1); the partial sums converge to $\{\Delta, 6\ln 3\}$ monotonically from below. *Coupling cross-link [E] (audit):* the established seam coupling is $\kappa = \frac{c}{24\pi} = \frac{8}{24\pi} = \frac{1}{3\pi} = \alpha_1/\pi$ — the *first parabolic weight over* π. *The wall [E]:* the ring series diverges at $\alpha \to 1$ ($n \to N_{\rm fam}$) — the v124 wall is the standard RPA instability. *R1 final form [C]:* show that the near-Nariai one-loop effective action of the seam mode, with insertion = parabolic weight and multiplicity p_2, is this ring sum — "**the seam clock is the RPA determinant of one mode on six sites**". The resummation type is identified; the derivation itself remains open.

Where the six sites live ([verification/v128_graded_hull.py]): the Ginsparg–Perry $l = 1$ sector — the

zero modes $\lambda \in \{0, -1\}$ of the classical clock (v104/v107) — has $2l+1 = 3$ modes, and sheet doubling gives $2 \times 3 = 6 = p_2$ [E]: the ring multiplicity equals the sheet-doubled zero-mode count. [C] reading (recorded): zero modes are exactly the modes that force nonperturbative resummation in de Sitter one-loop physics — the natural home of the six RPA towers. [verification/v127_ring_resummation.py]

The clock is an entropy power law ([E]/[C]). What the resummation argument $1 - n/N_{\text{fam}}$ *is* physically: the three established Nariai entropy levels (v101) are $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ (pure dS; two-horizon total; single horizon — deficit $S_{dS}/3$ per step) — exactly the complementary cusp weights. Hence [E]:

$$\boxed{\lambda_n = \left(\frac{S_n}{S_{dS}}\right)^{p_2}, \qquad \text{rate}(n) = p_2 \ln \frac{S_{dS}}{S_n}}$$

— the transfer eigenvalues are the *entropy fractions to the hexagon power*; the rates are hexagon-weighted log-entropy deficits. *The coupling is the entropy deficit [E] (triple identity):* $\alpha = n/N_{\text{fam}} =$ the Mehta–Seshadri parabolic weight (v115/v126) $= 1 - S_n/S_{dS} = \Delta S_n/S_{dS}$ — the RPA insertion strength is the fractional entropy deficit; the coupling is not a free parameter in *any* of its three readings. *Orientation [E]:* weight $0 \leftrightarrow$ pure dS (maximal entropy, stationary attractor), weight $2 \leftrightarrow$ single horizon (fastest decay) — the established repeller→attractor flow. *R1, thermodynamic form [C]:* derive $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ from the near-Nariai one-loop — a Gibbons–Hawking-type entropy power law with exponent = the hexagon size (audit: $p_2 = 6 = 2h$ reads as the two-point form of a weight $h = 3 = N_{\text{fam}}$ operator; recorded, not claimed).

The Born square ([verification/v130_born_square.py]): the weight audit is now *derived* from mode counting [E]. In the standard collective-coordinate treatment each zero mode contributes a moduli factor $\propto S^{1/2}$, so with $n_{\text{zero}} = 6$ the transition *amplitude* scales as $A \sim (S/S_{dS})^3$ and the *probability* is its Born square $\Gamma = |A|^2 = (S/S_{dS})^6$:

$$\boxed{h = \tfrac{n_{\text{zero}}}{2} = 3 = N_{\text{fam}}, \qquad p_2 = 2h \text{ (the Born rule)}}$$

— and the 3 is the dimension of the l=1 eigenspace = the number of *proper conformal Killing vectors of* S^2 (the classic near-Nariai zero modes, Ginsparg–Perry/Bousso–Hawking class), while the $\times 2$ is the *horizon pair* = the SdS mass-line deck (v101). The same 3 counts the generations and the sphere's conformal directions. *Residue [C]:* justify the per-zero-mode $S^{1/2}$ normalisation on the Nariai instanton for exactly these six modes — a textbook-class computation; no exotic object remains in R1.

The measure is the area ([verification/v131_measure_is_area.py]): the geometric core of that normalisation is exact [E]: on $S^2(r)$ the L^2 norm of *each* of the three normalised l=1 harmonics is

$$\boxed{\|Y_{1m}\|^2 = r^2 = \frac{A}{4\pi}}$$

(all three integrated exactly) — the zero-mode norm *is* the horizon area, hence the entropy. The collective-coordinate Jacobian per mode is the mode norm $\sim S^{1/2}$ (standard change of variables), so: norm = area → Jacobian = $S^{1/2}$ → six modes → amplitude S^3 → Born → S^6 — every exponent factor carries a derivation-level origin. The 4π is the same Gauss–Bonnet primitive that builds $c_3 = \frac{1}{2}(4\pi)^{-1}$, and it *cancels* in all ratios — exactly why the clock sees only entropy fractions. *The last step [C]:* show $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$ (non-zero-mode determinant cancellation; same local near-Nariai geometry). [verification/v129_entropy_power_law.py]

The cancellation, exact within the family ([verification/v144_detratio_family.py]): that step is now *derived* in the within-family sense [E]. The traceless SdS cubic forces

$$\boxed{r_b\, r_c = 1 - \frac{\Delta^2}{3} \quad (\Delta = r_b - r_c \text{ the horizon split; exact, no higher corrections})}$$

(e_2-rigidity), and with the v132 anomaly $\zeta(0)|_{\det'} = -\frac{2}{3}$ per sphere the two-sphere non-zero-mode determinant ratio against Nariai is the closed form $\det'_n / \det'_N = (1 - \Delta^2/3)^{4/3}$: **no first-order term in the split** — the hoped-for cancellation is a consequence of e_2-rigidity plus

the established anomaly, not an assumption. In deficit form, $\varepsilon = 1 - c = \frac{3}{8}\Delta^2 + O(\Delta^4)$ gives $\det'_n / \det'_N = 1 - \frac{32}{27}\varepsilon + O(\varepsilon^2)$ with $\frac{32}{27} = |\mu_4|(\frac{|\mathbb{Z}_2|}{N_{\rm fam}})^3$ (the v103 canonical curvature times the glue order; audit-typed). *Honest scope [C]:* the finite-weight comparison ($S_n/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$) still runs through the v132/v133 budget — a naive global-rescaling reading would shift the Born exponent $6 \to \frac{14}{3}$ (not an atom), so the absorption must proceed via the Hubble-unit conversion; the frozen transfer spectrum is untouched, and the remaining [C] now has its obstruction stated sharply.

The clock is a Gaussian zero-mode integral ([verification/v147_clock_gaussian.py]): the v127 typing job is discharged at the model level, and the bend is $\det'$-clean [E]. For p_2 Gaussian zero modes whose per-mode variance scales by $(1-\alpha)$ — *forced* by norm = area (v131) — the Born-squared partition ratio is exactly $(1-\alpha)^{p_2}$, i.e. rate $= -p_2 \ln(1-\alpha)$, and the ln series *is* the ring sum $p_2 \sum_k \alpha^k/k$ term by term. Sharper still: the two quantum weights $n = 1, 2$ are bookkeeping levels of *one* Euclidean geometry (Nariai $S^2 \times S^2$, v101), so $\det'_1 = \det'_2$ *identically* — **the bend $\log_{3/2} 3$ (the v107 target) carries no determinant correction at all**. The single across-topology step is the $\alpha = 0$ reference (round dS vs Nariai), exactly where the v132/v133 budgets and the v144 obstruction live; its classical side is already matched (v126: slope $2 = |\mathbb{Z}_2|$). *What stays [C]:* the measure identification ("the gravitational near-Nariai zero-mode measure *is* this Gaussian model") and the reference normalisation — structure, spectrum, bend and ring typing are exact.

The det-ratio anomaly is the Koide constant ([E]/[C]). The det-ratio question acquires its first *exact* invariant. The Ginsparg–Perry sphere operator on the unit S^2 is $L = -\Delta - 2$ (the $-2 = -|\mathbb{Z}_2|$ is the v104 modulus mass), with eigenvalues $(l+2)(l-1)$ and kernel the three $l{=}1$ zero modes. Its heat coefficient is made of atoms [E]: $a_1 = \frac{R}{6} - E = \frac{1}{3} + 2 = \frac{1}{N_{\rm fam}} + |\mathbb{Z}_2| = \frac{7}{3}$, and the scaling anomaly of the non-zero-mode determinant is

$$\boxed{\zeta(0)\big|_{\det'} = a_1 - \dim\ker = \tfrac{7}{3} - 3 = -\tfrac{2}{3} = -\frac{|\mathbb{Z}_2|}{N_{\rm fam}}}$$

— confirmed independently by numerical continuation of the heat trace ($\sim 10^{-7}$): **the det-ratio anomaly is the Koide constant**, the eighth appearance of 2/3 and the first as a *spectral* anomaly. *Consequence [E]:* $\det'(cL) = c^{\zeta(0)} \det'(L)$ — the determinant ratio between configurations is *not* automatically 1; the S^2-sector share of the scaling budget is exactly $-\frac{2}{3}$ per Hubble-unit conversion, while the dimensionless tower itself is configuration-independent (pure-number GP spectrum). *The budget statement [C] (the remaining step, made precise):* show the *full* $\zeta(0)$ budget of the 4d Nariai fluctuation problem (dS_2 + gauge/ghost + this S^2 sector) vanishes or is absorbed into the established power law — the S^2 share is pinned exactly; nothing is claimed about the total. [verification/v132_detratio_anomaly.py]

The zeta budget: the reduced route carries the anchor ratios ([E]/[C]). Euclidean Nariai is $S^2 \times S^2$ — the temporal sector continues to the *same* unit sphere, so both candidate determinant routes are exactly computable. *(i) The reduced route [E]:* each 2d sector carries the v132 operator with $\zeta(0)|_{\det'} = -\frac{2}{3}$; the two sectors give

$$\boxed{\text{budget}_{\rm red} = -\tfrac{2}{3} - \tfrac{2}{3} = -\tfrac{4}{3} = -\frac{e_1}{p_0} \text{ (minus the seed gain)}}$$

— the per-sector share and the total are *two of the three v119 anchor-ratio triad values*; the six zero modes split $3 + 3$ across the two spheres: the sheet doubling *is* the two-sphere structure of Euclidean Nariai (= the horizon pair, v101). *(ii) The naive 4d route [E]:* with the standard S^2 heat expansion $1/t + \frac{1}{3} + t/15$ and the symmetric split, $a_2(4d) = (\frac{4}{3})^2 + 2 \cdot \frac{9}{10} = \frac{161}{45}$, six zero modes, $\zeta(0)|_{\det',4d} = \frac{161}{45} - 6 = -\frac{109}{45}$ — *no* anchor-atom reading. *(iii) Route discrimination [C]:* the budget computation structurally favours the **reduced seam reading** (pure anchor-ratio anomalies), consistent with the theory's 2d-seam architecture throughout — route-selection evidence, not proof. *Remaining [C]:* the graviton/ghost shares on $S^2 \times S^2$ (Volkov–Wipf class, external standard) — a finite list of known heat coefficients. [verification/v133_zeta_budget.py]

The Volkov–Wipf firewall: the external edge is two copies of the reduced budget ([E]/[C]). The "finite list" above is now *filled in externally.* For the full one-loop graviton path integral on $S^2 \times S^2$ the log-coefficient is (arXiv:2506.02142, Table 4.1, agreeing with Volkov–Wipf, Nucl. Phys. B582 (2000) 313):

$$\alpha_{\rm bulk} = \frac{22}{45}, \qquad \alpha_{\rm edge} = -\frac{8}{3}, \qquad \alpha_{\rm PI} = -\frac{98}{45}$$

(no anchor atom; the difference to the reduced budget is $-\frac{38}{45}$ with $38 = 2 \cdot 19$). The structural finding: the external *edge* factor consists of **two identical copies** (one per horizon), and

$$\boxed{\alpha_{\rm edge} = -\frac{8}{3} = 2 \times \left(-\frac{4}{3}\right)}$$

— the per-copy magnitude $\frac{4}{3} = e_1/p_0$ is *exactly* the reduced seam budget above (the seed gain; $-\frac{2}{3}$ per sector). The TFPT reduced route lands on the **edge sector** of the external bulk/edge split; the bulk ($\frac{22}{45}$) is the ideal graviton gas, which the seam clock does not contain; two copies = the horizon pair = the sheet doubling (v101; the 3+3 zero-mode split). *Typing decision [C] (recorded):* R1 = the reduced zero-mode/RPA seam clock = an edge-sector object; the full 4d Volkov–Wipf determinant is the *gravity firewall* (its $-\frac{98}{45}$ is absorbed in the $\mu_0/(\Lambda G)^3$ normalisation), not a TFPT atom. Per-copy sign/orientation (the edge factor enters inverted) and the Lorentzian edge-mode reading stay [C]. [verification/v138_vw_firewall.py]

Black-hole mechanics in seam units ($c_3 = 1/(8\pi)$): every coefficient is a compiler atom

quantity	standard form	seam form	compiler atom
Einstein equations	$G_{\mu\nu} = 8\pi T_{\mu\nu}$	$G_{\mu\nu} = T_{\mu\nu}/c_3$	$c_3 = 1/(8\pi)$ (P1)
first law	$dM = \frac{\kappa}{8\pi} dA$	$dM = c_3\,\kappa\, dA$	seam constant
Smarr (Schw.)	$M = \frac{\kappa A}{4\pi}$	$M = 2c_3\,\kappa A$	
temperature	$T_H = \frac{\kappa}{2\pi}$	$T_H = 4c_3\,\kappa$	$1/(2\pi) = 4c_3$
entropy	$S = \frac{A}{4}$	$S = 2\pi c_3 A$	$1/4 = 1/\lvert\mu_4\rvert$
Bekenstein bound	$S \le 2\pi ER$	$S \le ER/(4c_3)$	
Hawking power	$P = \frac{1}{15360\,\pi M^2}$	$P = \frac{c_3}{1920\,M^2}$	$1920 = \lvert W(D_5)\rvert$
lifetime	$\tau = 5120\,\pi M^3$	$\tau = 128\,g_{\rm cat} M^3/c_3$	$128 = 2^7$, $7 =$ scalaron
Kerr extremal	$A_{\rm ext} = 8\pi M^2$	$A_{\rm ext} = M^2/c_3$	$A_{\rm ext}/A_{\rm Schw} = 1/\lvert\mathbb{Z}_2\rvert$
area quantum (Hod)	$\Delta A = 4\ln 3$	$\Delta A = \ln(N_{\rm fam}^4)$	$81 =$ disc of the cover
Nariai bound	$S_N = \frac{2\pi}{\Lambda}$	$S_N = \frac{2}{3} S_{dS}$	$2/3 = \lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ (Koide)

Figure 6: Black-hole mechanics in seam units: every classical coefficient is a compiler atom. [verification/v101_horizon_anchor.py]

6 Turnaround radius, gravitational waves, light

Bound structures. In a Λ-universe the maximal bound radius is $r_{\rm ta}(M) = \left(GM[\,2\pi e^{\alpha_\star^{-1}}/\bar{M}_{\rm Pl}]^2\right)^{1/3}$ — halo, cluster, supercluster and void scales follow from the Λ closure with no new parameter. [C]

Gravitational waves. The low-curvature branch is $R + R^2$; the tensor graviton runs on the Lorentz cone and the only extra mode (the scalaron, $M_{\rm scal} \approx 3.06 \times 10^{13}$ GeV) is frozen for astrophysical frequencies. Hence

$$\boxed{v_{\rm GW} = c} \qquad \text{(no measurable dispersion)} \qquad \text{[C]},$$

a clean falsifier: a genuine $v_{\rm GW} \neq c$ would break this gravity closure.

Light. TFPT does not predict the SI number c (definitional); it predicts the *universality of the causal cone* — RP → OS reconstruction → one Lorentzian cone shared by EM, gravity and massless modes ($v_\gamma = v_g = c$, no native photon dispersion). What it *does* compute is not a speed shift but a polarisation rotation, cosmic birefringence

$$\boxed{\beta_{\rm rad} = \frac{u}{4\pi} \approx 0.2424^\circ} \qquad \text{[E]},$$

from the same seed $u = \phi_0^{\rm ret}$ that fixes the Cabibbo angle, Ω_b and θ_{13}.

7 The local sibling: an EHT achromatic polarization intercept

The cosmic birefringence $\beta_{\rm rad}$ is the *asymptotic* (Thomson-limit) projection of the determinant-line response; the *local* projection onto the magnetised collar of a horizon-scale black hole is a structured, achromatic residual rotation of the linear-polarization angle [verification/v203_eht_achromatic.py]

$$\beta_{\rm BH}(r) = \frac{Q_e^{\rm eff}(x)\, Q_m^{\rm eff}(x)}{256\pi^4\, r^2} = 16\, c_3^4\, \frac{Q_e^{\rm eff} Q_m^{\rm eff}}{r^2} = \frac{\delta_{\rm top}}{3}\, \frac{Q_e^{\rm eff} Q_m^{\rm eff}}{r^2},$$

where the coupling $16c_3^4 = 1/(256\pi^4)$ is fixed *exactly* ([E]) — the *same* top-form coefficient $\delta_{\rm top} = 48c_3^4 = 3/(256\pi^4)$ that controls the α-kernel precision-zone correction. There is *no free coupling*: EHT-class polarimetry probes the same admissibility data as the α row, only projected onto the local collar. What TFPT fixes is the structured $1/r^2$ shape, the achromaticity, and the sign-flip under effective $E{\cdot}B$ reversal, *not* the amplitude $Q_e^{\rm eff} Q_m^{\rm eff}$ (an MHD/GR weight); the channel is therefore typed [C] (shape/sign), not a parameter-free pixel prediction.

Dated kill test [X]. Build the residual intercept $\chi_0^{\rm res} = \chi_0^{\rm obs} - \chi_0^{\rm GRMHD}$ after honest GRMHD/synchrotron subtraction; it must pass *three independent nulls* — frequency (achromatic, no λ^2 tail), spatial ($1/r^2$ profile), and sign-flip (reverses under $E{\cdot}B$ reversal). A residual statistically consistent with zero across the horizon-scale image, or failing any null, falsifies *this channel* (the compiler core is untouched). *Scope:* the old UFE closed RN-type metric (D/r^4, regular cores, shadow shift) is *not* adopted — it is superseded by the Nariai/seam=horizon readouts above (v190); only the polarization signature survives.

8 Search targets (not claims)

Audit-level search ansätze [O]

- **Black-hole echoes / horizonless compactness:** any near-horizon echo amplitude ratio would be a natural transfer-rate target, $\mathcal{A}_{n+1}/\mathcal{A}_n \lesssim (2/3)^6 \approx 0.0878$. The *same* window $Q_{\rm geom} \in [\frac{3}{8}, \frac{1}{2}]$ that the SdS roots trace surfaces independently as the maximum horizonless gravastar compactness $\mathcal{C} = \frac{3}{8}$ (de Sitter endpoint $\frac{1}{2}$) of Jampolski–Rezzolla (2026); since $\frac{1}{3} < \frac{3}{8} < \frac{4}{9} < \frac{1}{2}$ such an object is a horizonless light-trapping ECO. `experiments/gravastar-compactness` turns this into an echo template (round-trip delay ~ 0.7 ms at $62\,M_\odot$, amplitude $\leq (2/3)^6$) — a [C] structural echo, no $\mathcal{C} \leftrightarrow Q_{\rm geom}$ map proven. A search ansatz, no prediction.
- **Quantum recovery:** the Page-curve recovery kernel $I \sim (2/3)^{6n}$ is a falsifiable shape if a boundary-recovery rate is ever measured.
- **BH HFQPO ladder tooth (×1.5):** the four published black-hole HFQPO twin pairs are consis-

tent with exactly 3:2, but that cluster is *cheap* (an anchored detection-selection null reproduces it in 18.5% of trials, and XTE J1859+226 breaks universality at $+9.2\sigma$), and mapping the TFPT relaxation-ladder step 3/2 (from $\mathrm{rate}(n) = -6\ln(1 - n/N_{\mathrm{fam}})$, $N_{\mathrm{fam}} = 3$) onto a two-oscillator frequency ratio is *non-canonical* — GR parametric resonance is the standing favorite. The one discriminating target is a *third tooth* at $\nu_3 = (3/2)\,\nu_u$ (661.5/414/252/363 Hz for the four sources) with integer harmonics forbidden on the geometric ladder. No published search ever targeted these frequencies; the preregistered archival RXTE PCA scan (executed 2026-07 on 77 ObsIDs across the four sources; sanity gate 11/12 published pair lines reproduced, injection-calibrated in every source) finds *neither* the tooth *nor* the integer control line $4\nu_0$ anywhere ($\sim 0\sigma$, trials-corrected), with 3σ rms upper limits of 0.53–3.06% — in every source at or below the strength of the detected upper pair line. A well-powered null: the ladder reading gains no support (GR resonance stays favored) but is not killed, since the harmonic branch did not fire either (the published integer lines 92=184/2, 34/68 Hz stand). `experiments/hfqpo-ladder` carries the preregistered census, selection-null Monte Carlo and the executed archival scan. A search ansatz, no prediction; even a future tooth hit would be [C] until the mapping is derived.

- **FRB repeaters:** fast-radio-burst repeaters are a *preregistered search interface* for the same boundary-recovery kernel (the frozen ratios $\{1, (2/3)^6, (1/3)^6\}$ and $\{1, (2/3)^3, (1/3)^3\}$), *not* a horizon claim and *not* part of the verification ledger — the work lives in `experiments/frb-tfpt-signatures` with a frozen hypothesis file and surrogate-calibrated null tests; current public data neither confirm nor refute the kernel (a clean null on echo ratios, a single non-replicated window candidate). Hunting ground, not foundation. A joint five-year *power forecast* across the comb channels (PSR/FRB/GW/UHECR/CMB and siblings) now lives in `experiments/p9-forecast` (preregistered before any Monte-Carlo run, anchored to the published sibling calibrations; verdict P9-FORECAST-GO on two channels plus the hierarchical combination, with its three load-bearing conditionals named) — a forecast, *no* data analysed, *no* evidence row, firewalled exactly like the siblings.
- **Cosmological coupling ($w = -1$):** if the de Sitter–vacuum seam interior (the regular-core reading above) makes a seam black hole carry the Λ vacuum, $w_{\mathrm{in}} = -1$, then under Croker–Weiner cosmological coupling $k = -3w_{\mathrm{in}}$ the mass-growth index is $k = 3$ exactly, i.e. a population $w = -1$. `experiments/ccbh-dark-energy` confronts this with Farrah+2023 ($k = 3.11 \pm 0.79$, -0.14σ) as a *contested* downstream bridge and an *alternative reading* of the same $w = -1$ the DESI w-watchdog tests — not two independent hits.
- **Cosmic spin parity (frontier):** the μ_4 clock and the $PSL(2,\mathbb{C}) \cong SO^+(3,1)$ boundary orientation allow at most a tiny global remnant, so TFPT predicts *approximate parity*, not a large handedness excess. `experiments/cosmic-handedness` watches the galaxy-spin asymmetry (Shamir JADES $\sim 3.3\sigma$ monopole) — most likely a Milky-Way-rotation *dipole* systematic and isotropic in Galaxy Zoo — a frontier watchdog, explicitly *not* promoted.

Net

All horizon formulas share the one seam normalisation $1/(2\pi) = 4c_3$. Black-hole, de Sitter and Unruh temperatures, the Page time, scrambling, the Nariai bound, turnaround radii, $v_{\mathrm{GW}} = c$ and birefringence are *readouts of one seam constant* — standard physics rewritten in TFPT units, with two genuine compiler fingerprints (1920 = $|W(D_5)|$ in Hawking power, $|\mu_4| = 4$ in scrambling). No new gravitational physics is claimed; the value of the reframe is that horizons stop being separate modules.

Appendix: computational verification

The seam-unit identities of this note (the thermal grammar $\frac{1}{2\pi}$=$4c_3$, the power denominator 1920=$|W(D_5)|$, the de Sitter entropy form, the Page fraction, the scrambling $|\mu_4|$=4, and β_{rad}=$\varphi_0^{\mathrm{ret}}/4\pi$=0.2424°) are re-derived and machine-checked in `verification/v8_horizon.py`. Run `python verification/v8_horizon.py` (needs `mpmath`); it imports the same c_3=$1/(8\pi)$ primitive as the rest of the suite. The search-target ansätze remain [O] and are deliberately not part of the pass/fail checks.

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

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- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
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